Aarush Mehrotra

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Education

2026 San Diego, USA — **B.S. Mathematics-Computer Science,** *University of California, San Diego*
Relevant coursework: Advanced Data Structures, Recommender Systems, Software Engineering, Data Science, Honors Linear Algebra, Computer Organization, Econometrics, Math of Modern Cryptography, Directed Group Study (MLPerf/BERT/SDXL, HPL, ICON on multi-node clusters)

2023 — **Python Data Products for Predictive Analytics Specialization,** *University of California, San Diego (online)*
Applications in preprocessing, pipelines, and generative predictive analytics in a wide range of fields, supplemented by project and competition work.

Skills

AI / ML: PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, NumPy, MLPerf benchmarking, LLM architecture, Inference optimization, AMD Instinct GPU workloads (HIP, ROCm), CUDA, Random Forest, Gradient Boosting, Neural Networks, Deep Learning, Feature Engineering, Time-Series modeling.

Systems & Infrastructure: OpenMP, MPI, Slurm, Docker, Singularity, Kubernetes, Ansible, Spack, Conda, Kokkos, Warewulf, iDRAC/IPMI, PXE-based OS installation, provisioning, and troubleshooting.

Professional Experience

San Diego Supercomputer Center

02/2025 – 03/2026 San Diego, CA — *HPC Intern*
- Delivered 100% accessibility score and tag-based searchability on the Jupyter Notebook Hub; presented results as first-author poster at JupyterCon 2025.
- Sole developer and owner of multiple Jekyll sites deployed via GitHub Actions with branch-protection rules and automated deployment workflows supporting external contributors.
- Data pipeline development for internal LLM training and inference use.
- Maintained and improved tutorials on the Expanse HPC cluster; documented dependency pipelines.

04/2024 – 06/2024 — *Student Software Engineer Intern*
- Ported and optimized Fortran HPC codes for earthquake modeling with Oak Ridge National Lab.
- Debugged and accelerated simulation throughput and added a checkpointing feature.
- Collaborated with Ohio State faculty to enhance GPU-aware compiler performance (MVAPICH2).

07/2023 Remote, USA — **Trinitas Capital,** *Intern*
- Built an AI-driven trading signal system in one month using historical price data.
- Developed neural network-based buy/sell signal system using feature extraction on historical price data, achieving a 13% model return, outperforming the underlying asset by 10%.

Projects

2023 – 2025 — **Student Cluster Competition,** *SC24, Atlanta, GA*
- Led ICON project team of 4, optimizing weather simulations and automated visualizations in a time- and resource-constrained environment.
- Set up user accounts (Ansible), package managers for team-wide dependencies (Spack), set up Slurm.
- Contributed to MLPerf benchmarks on AMD GPUs, optimizing PyTorch backends for SDXL.
- Supported sponsor outreach and peer mentorship as a senior team member, including in 2025.

2025 — **Single Board Cluster Competition,** *San Diego Supercomputer Center, UCSD*
- Led hardware selection and OS/network setup for a winning HPL benchmark run.
- Contributed to MLPerf runs, securing first place in overall competition.

Publications

2025 — **Poster: Taggable, Accessible Jupyter Notebook Hub for HPC Training,** *JupyterCon 2025*

2024 — **Preliminary Results of the MLPerf BERT Inference Benchmark on AMD Instinct GPUs,** *ACM (PEARC '24)*
A paper about ML inference optimization and model performance at scale on specific AMD systems.